

Chemistry Lab

Answer Key

1. A
2. D
3. D
4. C
5. C
6. A
7. A
8. C
9. C
10. B
11. C
12. B
13. A
14. C
15. C

16.
 - (a)
 - 1: high pressure
 - 1: low temperature
 - (b)
 - 1: the a value of methane is lower
 - 2: CH_4 has weaker intermolecular forces than does H_2O because CH_4 is nonpolar while H_2O is polar (has hydrogen bonding)
 - (c)
 - 1: the b value relates to particle volume (can be stated or implied)
 - 2: atomic radius increases down a group due to increased electron shielding (additional energy levels)

17.
 - (a) $\text{Al}_2\text{S}_3 (s) + 6\text{H}_2\text{O} (l) \rightarrow 2\text{Al}(\text{OH})_3 (s) + 3\text{H}_2\text{S} (g)$
 - 1: chemical formulas and state symbols
 - 1: balanced equation (coefficients may be in multiples)
 - (b)
 - 1: $4.04 \text{ atm} - 0.970 \text{ atm} = 3.07 \text{ atm}$
 - 2: substitution into ideal gas law
 - 1: molar mass of Al_2S_3 conversion
 - 1: 94.1 %

- 18.**
- (a)**
- 1: $k_{2f}[\text{Cl}][\text{CO}] = \text{Rate}$
 - 1: $k_{1f}[\text{Cl}_2] = k_{1r}[\text{Cl}]^2$
 - 1: $k[\text{Cl}_2]^{1/2}[\text{CO}] = \text{Rate}$ (concentrations may instead be expressed as partial pressures)
- (b)**
- 1: $k(4.0)^{1/2}(1.0) = 6.0 \times 10^{-3}$
 - $k = 3.0 \times 10^{-3} \text{ M}^{-1/2} \text{ s}^{-1}$
 - 1: value of k
 - 1: units of k
- 19.**
- (a)**
- 2: breaking the dry ice block into small pieces increases total surface area (amount of dry ice molecules exposed)
 - 1: increased number of collisions per second with (presumably) warmer atmospheric particles
- (b)**
- 2: catalysts provide an alternate reaction pathway with lower activation energy
 - 1: greater fraction of collisions have enough kinetic energy to overcome activation energy
 - Alternate 1: by the Arrhenius Equation a lower E_a corresponds to a higher k
- (c)**
- 2: proper orientation of reactant molecules is required to reach transition state
 - 1: in most cases the N atom in NO must collide with an O atom in O_3 to form NO_2
- 20.**
- (a)**
- Possible Methods: (1) measure change in pH directly using pH meter as OH^- reacts; (2) use two pieces of litmus paper to measure change in pH at two distinct times; (3) use spectrophotometer to measure change in indicator (phenolphthalein, thymol blue) absorption as OH^- reacts; (4) use AgNO_3 to precipitate AgBr as Br^- forms and measure the mass of AgBr produced
- 1: mention of rate of change of OH^- and / or Br^- (can be stated or implied)
 - 2: first method procedure and explanation
 - 2: second method procedure and explanation (must be different than first)
- (b)**
- 1: calculation of slope to determine average rate
 - 3: determination of linear plot by tests of logarithmic and inverse (if necessary) transformations
 - 1: first-order in bromoethane